INCH-POUND

MIL-PRF-81757/18(AS)
29 May 2009

PERFORMANCE SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, NICKEL-CADMIUM, HIGH RATE, 12-VOLT, 55-AMPERE-HOUR, LOW-MAINTENANCE, VENTED

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-81757.

REQUIREMENTS:

1. Part or identifying number. The part or identifying number (PIN) of the battery shall be M81757/18-1. The M81757/18-1 is a limited-repair, low-maintenance, vented battery that includes the removable M81757/18-2 sensor/harness assembly.

2. Dimensions and weight. The dimensions and weight of the battery shall be as specified in 3.10.8 as modified herein and as shown on figure 1 of MIL-PRF-81757/18(AS). The weight of the battery shall be not greater than 55.0 pounds (25.0 kg).

3. Capacity. The minimum capacity of the battery shall be as specified in 3.10.14 as modified herein.

4. Interface requirements between battery and test fixtures. The battery shall contain 10 cells. The cells and other interior components of the battery shall be arranged and positioned as shown in either layout of figure 2 of MIL-PRF-81757/18(AS). The fasteners specified on figure 2 are required due to interfaces with the J1 receptacle (MS3509-28), the J2 receptacle, and the sensor/harness assembly (M81757/18-2).

5. Interface requirements between the battery and the sensor assembly. The height of the partition at the front of the case shall be within ¼" of the cover-to-jar seal of the cell. The partition shall have rigidity and strength to contain the cells for the life of the battery without warping or breaking. The partition shall not distort due to undergoing any or all of the tests specified herein.

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6. M81757/18-2 sensor/harness assembly. Each battery shall have an M81757/18-2 sensor/harness assembly that shall contain two thermal switches and a voltage lead. The thermal switches shall be normally open; they shall close at 155 °F ± 5 °F and open within a 9 °F temperature drop from the closing temperature. The volt lead shall have a resistor of value not less than 1 K ohm ± 2 percent 1 watt for short circuit protection. The J-2 connector shall be a jam nut style receptacle that shall mate with an MS3116P128S plug. The voltage tap and thermal switches shall mount on to the intercell connectors as shown on figure 2 herein. The M81757/18-2 sensor/harness assembly shall conform to figure 3 of MIL-PRF-81757/18(AS).

7. MIL-PRF-81757 variance. The battery shall comply with MIL-PRF-81757 except as follows.

7.1 Modify the following paragraphs:

3.7.2 Battery container and cover. Delete the entire text and substitute: “The battery container, cover and any material used to protect the cover (such as paint, epoxy, or powdered coating) shall be made of a material that is impervious to salt spray and to an aqueous solution of potassium hydroxide of 1.30 ± 0.04 specific gravity (see 6.10.1 of MIL-PRF-81757) and an alternate electrolyte if used. Use of identical materials in the construction of the container and cover is not required as long as they meet the requirements of MIL-PRF-81757/18(AS). The cover shall be blue. Protective materials shall not be applied to the battery container except as indicated in 3.8.1 of MIL-PRF-81757. The battery container and cover shall be free of pits, weld blow holes, rough spots, and other deformations. The dimensions and locations of receptacles, hold-downs, latches (catches and strikes), and battery identification shall conform to figure 1 herein. The cover shall be fitted with a hold-down pad that functions as the cover’s gasket and shall have integral projections that hold down each battery cell. The cover pad/gasket shall be sized and fitted to match the rim of the container body. The location of the projections shall be determined by the location of each vented filler cap. The projections shall contact the entire top of each vented filler cap and apply a force on each vented filler cap to restrain vertical movement of the cells during vibration in accordance with 4.5.16 of MIL-PRF-81757. Metal or wood products (such as cork) shall not be used as the gasket. If elastomers are used as the gasket, the elastomer shall conform to the requirements of 3.10.3, as modified herein (see 6.10.2 of MIL-PRF-81757).”

3.7.5 Handles. Delete the title and entire text and substitute: "Carrying straps. The battery shall have two flexible carrying straps located as shown on figure 1 herein. The straps shall be a dark color and shall be free of any coating. The straps shall have a means by which they can be connected to each other over the top of the battery. Fasteners or adhesive methods shall not be used to attach the straps' mounting supports to the battery. The straps shall be nonremovable and shall be permanently anchored to the battery case at four separate locations with separation as shown in figure 1 herein. The straps shall not show any evidence of breaking away from the container or other damage after being tested in accordance with 4.5.9, as modified herein. Straps shall meet the requirements of 3.10.3 herein, and 3.9, 3.10.10, and 3.10.25 of MIL-PRF-81757."

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3.7.7 Vent tubes. Delete title and entire text and substitute the following: "Venting. The battery case shall contain vent(s) for release of gases, as depicted on figure 1 herein. Changes to the vent design shall be requested by the manufacturer in writing to the qualifying activity. Vent tubes shall not be used. Vents shall be located such that the container cover or internal battery components do not block the vents. The air inlet vents, which may be in the form of holes, louvers, etc, shall have an open area of not less than 2 square inches in the lid and not less than 2 square inches in the battery container. If any dimension of the vent(s) exceeds 0.25 inches, the vent(s) shall be covered by a means that prevents debris from entering the battery. When the battery is operating within the requirements of MIL-PRF-81757, cell venting shall not cause the battery to discharge entrained electrolyte."

3.7.8 Receptacles. Following the last sentence, insert: “The J1 receptacle shall conform to MS3509-32 (preferred) or MS3509-28. The receptacle shall be vertically positioned so that the top of the terminal block and the cell terminal contact surfaces are in the same plane. The J2 receptacle plating shall contain neither cadmium nor chromium.”

3.7.11.6 Vented filler cap. Following the first sentence, insert: "Each cell shall contain a vented filler cap conforming to PIN M81757/2-5 snorkel."

3.8.4.1 Limited-repair batteries. In the first sentence, change "figure 1" to "figure 4 of MIL-PRF-817857/18(AS)".

3.8.4.3 Battery caution and receptacle marking. In the second and third sentences, change "figure 3" to "figure 5 of MIL-PRF-81757/18(AS)". Insert the following after the third sentence: "The assigned CAGE code shall be inserted at the indicated area."

7.2 Add the following paragraph to MIL-PRF-81757:

"3.8.4.4 Caution label for matching battery modules. A caution label concerning matched batteries as shown in figure 6 of MIL-PRF-81757/18(AS) shall be attached to the right face of the battery, as seen from the receptacle. The marking shall be in accordance with figure 6 herein in terms of required information and sequence of information. The marking background shall be as large as possible yet not touch the area under the latches nor the area under the strap fastening mechanism. The marking shall use a lettering height of not less than 0.25 ± 0.02 inch (for example, 18 point type or larger) for all wording except the "CAUTION", which shall use a lettering height of not less than 0.50 ± 0.02 inch (for example, 36 point type). The wording shall be in white print on a blue background (color range 15090 to 15193 of FED-STD-595)."

7.3 Modify the following paragraphs of MIL-PRF-81757:

3.8.5 Cell identification. Following the last sentence, insert: "Cells procured as replacement cells shall have the positive half of the cover (cell lid) colored red. Cells provided as original components of the battery shall have the entire cover (cell lid) colored the natural color of the material."

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3.10.3 Electrolyte resistance (components and elastomeric materials). Delete the title and the entire text and substitute: "Sensor/harness assembly, components, and elastomeric materials. The manufacturer shall provide three samples of the M81757/18-2 sensor/harness assembly in addition to the samples of 4.2.2 of MIL-PRF-81757."

7.4 Add the following paragraphs to MIL-PRF-81757:

"3.10.3.1 Effects of test. The sensor/harness assembly, components, and elastomeric materials shall meet the requirements of 3.9 of MIL-PRF-81757 and the requirements shown below during and after testing in accordance with 4.5.3, as modified herein.

"a) The J2 receptacle, components (see 6.13.3 of MIL-PRF-81757) and elastomeric materials used in the battery shall not exhibit cracks, blisters, or other deleterious effects.

Affected part of the assembly	Deleterious effects
External and internal parts	Cracking, pitting, chipping, scaling, blistering, corrosion (however, if a component [for example, a spring or receptacle pin] is made of 400-series stainless steel, then minor surface corrosion is permitted), or other deleterious effects
Electrical parts	Functional degradation
Plastic materials	Change more than 2.5 percent in weight or 2.0 percent in any major dimension (length, width, or height)
Identification markings	Smudging, smearing, chipping, crazing, or other deleterious effects

"b) The color and identification marking of the J2 receptacle shall be impervious to the test liquid.

"c) Electrical parts shall perform with no evidence of functional degradation.

"d) All the components of the sensor/harness assembly, including thermal switches, transducers, and internal heaters, shall meet the requirements specified herein.

"3.10.3.2 Switch activation. The sensor/harness assembly shall perform as follows during testing in accordance with 4.5.3.1.a and 4.5.3.1.e of MIL-PRF-81757/18(AS). The thermal switch contacts shall close at 155 °F ± 5 °F on a temperature rise and open within a 7 °F ± 2 °F temperature drop from the closing temperature.

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"3.10.3.3 Switch contact resistance and dielectric resistance. The sensor/harness assembly shall perform as follows after being tested in accordance with 4.5.3.1.a and 4.5.3.1.d, both of MIL-PRF-81757/18(AS). The resistance between pin G and the voltage tap terminal lug shall be 1 Kohm ± 2 percent. The resistance between the following points shall be (1) not less than one megohm at temperatures less than 150 degrees of, and (2) shall be not greater than 2 ohms at temperatures above 160 degrees of:

"a. Pins C and E.
"b. Pin C and the case.
"c. Pin E and the case.
"d. Pin G and the case.

"The J2 receptacle shall also meet the requirements of 3.8.1, 3.8.4, and 3.9, all of MIL-PRF-81757."

7.5 Modify the following paragraphs of MIL-PRF-81757:

3.10.8 Dimensions and weights. Delete the entire text and substitute: "The dimensions and configuration of the battery container and container cover shall be as shown on figure 1 herein. The weight of the battery shall be not greater than 55 pounds (25.0 kg) after the inspection of 4.5.8 of MIL-PRF-81757."

3.10.11 Cover seal. Delete the entire paragraph.

3.10.14 Capacity. Delete the first sentence and insert: "The minimum capacity (see 6.9, 6.13.1, 6.13.2, and 6.13.14, all of MIL-PRF-81757) of the battery shall be not less than 55 Ah. C-rate discharges shall be conducted at 55 amperes."

3.10.14.1 Initial capacity. Delete the entire text and substitute: “To meet the initial capacity requirement, each cell shall produce a potential of not less than 1.0 volt at not less than 75 minutes into the constant 55-ampere discharge after being tested in accordance with 4.5.13 of MIL-PRF-81757 at room temperature (see 4.4.1 of MIL-PRF-81757).”

3.10.14.2.2 Other batteries. Delete the entire paragraph.

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3.10.22 Constant voltage discharge. Delete the entire text and substitute: "The battery shall produce not less than the current specified below after being tested in accordance with 4.5.22 of MIL-PRF-81757. The battery shall also meet the requirements of 3.9 of MIL-PRF-81757 during and after charge and discharge."

Temperature	Average current over the total 60 seconds	Current at 15 seconds
75.2 °F ± 3.6 °F (24.0 °C ± 2.0 °C)	950 A	1000 A
-14.8 °F ± 3.6 °F (-26.0 °C± 2.0 °C)	500 A	500 A

3.10.23 Cycling. Delete the entire text and insert "Batteries shall meet the following requirements when cycled (see 6.13.5 of MIL-PRF-81757) in accordance with 4.5.23, as modified by MIL-PRF-81757/18(AS) under the ambient conditions of 4.4.1 of MIL-PRF-81757.

"a. The battery shall produce a potential of not less than 10.5 volts at the end of each 4.0-minute duty cycle discharge.

"b. The battery shall produce not less than 53 Ah when discharged in accordance with 4.5.23.c as modified by MIL-PRF-81757/18(AS). During group C inspection, each cell shall produce a potential of not less than 0.95 volts at not less than 60 minutes into the room temperature discharge of 4.5.23.c, as modified herein.

"c. The battery shall successfully complete a total of 700 duty cycles for qualification inspection and a total of 200 duty cycles for group C inspection.

"d. The battery shall also meet the requirements of 3.9 of MIL-PRF-81757 during and after charge and discharge."

VERIFICATION

1. MIL-PRF-81757 variance. The battery shall comply with MIL-PRF-81757 except as follows:

1.1 Add the following paragraph:

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"4.4.1.1 Temperature measurement location. When the battery temperature is to be measured during laboratory testing, the temperature shall be measured at the location shown on figure 2 herein.. The temperature shall be measured with a temperature sensor (thermocouple or thermistor) inserted to contact the top of the partition."

1.2 Modify the following paragraphs of MIL-PRF-81757:

4.4.2 Constant potential charging method. In subparagraph a, delete "28.25 volts" and substitute: "14.88 volts".

4.4.3 Constant current charging method. Delete the third sentence and insert: "Constant current charge the battery at room temperature at 30A (± 3 percent) for 2.5 hours, then reduce current to 12A (± 3 percent) for an additional 2 hours (see 6.13.7 of MIL-PRF-81757)." In 4.4.3.a, delete the second sentence and substitute: "The only exceptions are the adjustments specified in 4.5.8, 4.5.22.a, 4.5.23.a(2), 4.5.24.a, 4.5.28.a, and 4.5.28.c(3)." Paragraphs 4.5.8, 4.5.24.a, 4.5.28.a and 4.5.28.c (3) are all as specified in MIL-PRF-81757. Paragraphs 4.5.22.a and 4.5.23.a (2) are as modified by MIL-PRF-81757/18(AS).

4.5.3 Electrolyte soak of components and elastomeric materials. Delete the title and entire text and substitute: "Sensor/harness assembly, components and elastomeric materials test. Subject the sensor/harness assembly to the test of 4.5.3.1 herein. Subject samples of all other components (see 6.13.3 of MIL-PRF-81757) and materials of the cells and batteries and the backshell of any receptacle (other than J2) that is inside the battery case to the test of 4.5.3.2, as specified by MIL-PRF-81757/18(AS)."

1.3 Add the following paragraph to MIL-PRF-81757:

"4.5.3.1 Sensor/harness assembly test. Obtain an M81757/18-2 sensor/harness assembly. Do not disassemble the M81757/18-2 assembly. This test shall consist of the following steps.

"a. Stabilize an environmental chamber at 145 °F. Raise the temperature gradually and determine the temperature at which the switch contacts of the M81757/18-2 sensor/harness assembly close. Do not exceed 165 °F. Examine the assembly for the switch activation requirements of 3.10.3.2, as specified by MIL-PRF-81757/18(AS). Measure the resistance between pin G and the voltage tap terminal. Measure the resistance between pins C and E. Measure the resistance between pin C and the case. Measure the resistance between pin E and the case. Measure the resistance between pin G and the case. Examine the receptacle for the electrical resistance requirements of 3.10.3.3, as modified by MIL-PRF-81757/18(AS). When at this raised temperature, subject the sensor/harness assembly to a temperature drop. Determine the temperature at which the switch contacts open. Examine the assembly for the switch activation requirements of 3.10.3.2, as specified herein.

"b. Install the M81757/18-2 sensor/harness assembly on the inside of a container which is capable of withstanding 1.30 ± 0.04 specific gravity potassium hydroxide electrolyte. Attach the

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harness connector to the container in the same way it is attached to the battery. Fill the container with an aqueous solution of potassium hydroxide of 1.30 ± 0.04 specific gravity to cover all parts of the harness. Place the container in a 150 °F ± 3.6 °F (66 °C ± 2 °C) environment to bring the electrolyte solution to 150 °F ± 3.6 °F (66 °C ± 2 °C) and soak the assembly for 168 hours at that temperature.

"c. Rinse the specimen in water. Wipe dry. Examine the identification marking. Inspect all samples for evidence of deleterious effects. Examine the item for the requirements of 3.10.3.1, as specified in MIL-PRF-81757/18(AS) concerning color, identification marking, and lack of deleterious effects.

"d. Test the electrical resistance of the J2 receptacle at room temperature as follows. Measure the resistance between pin G and the voltage tap terminal. Measure the resistance between pins C and E. Measure the resistance between pin C and the case. Measure the resistance between pin E and the case. Measure the resistance between pin G and the case. Examine the receptacle for the electrical resistance requirements of 3.10.3.3, as modified herein.

"e . Clip the wires from the M81757/18-2 sensor/harness assembly. Perform the following on each switch separately. Stabilize an environmental chamber at 145 °F. Raise the temperature gradually and determine the temperature at which the switch contacts of the M81757/18-2 sensor/harness assembly close. Do not exceed 165 °F. When at this raised temperature, subject the sensor/harness assembly to a temperature drop. Determine the temperature at which the switch contacts open. Examine the assembly for the switch activation requirements of 3.10.3.2, as modified herein.

"4.5.3.2 Components and elastomeric materials test. Obtain three samples each of hold-down pads, gaskets, O-rings, and other items made of elastomeric material used in the battery or cell. Immerse the samples for 168 ± 2 hours in an aqueous solution of potassium hydroxide of 1.30 ± 0.04 specific gravity in a temperature chamber at 150 °F ± 2 °F (65.5 °C ± 1.1 °C). Rinse with water and wipe dry. Examine the specimens for the requirements of 3.10.3.1, as specified herein. Electrical components shall then undergo functional testing as indicated in MIL-PRF-81757/18(AS)."

1.4 Modify the following paragraphs, tables, and figures of MIL-PRF-81757:

4.5.9 Handle strength test. Delete the entire text and substitute: "Subject each strap individually to this test. Turn the battery case upside down on a table and permit the strap to hang over the side of the table. Hang 110 pounds of weight on the strap. Apply the weight continuously for not less than 5 minutes. Examine each strap and the battery for the requirements of 3.10.10 of MIL-PRF-81757."

4.5.14 Humidity and charge retention test. After the last sentence, insert "Recondition the battery in accordance with 4.4.5.2 of MIL-PRF-81757 before proceeding to the next test."

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4.5.22 Constant voltage discharge. Delete the text of subparagraph 4.5.22.a and insert "Constant current charge the battery in accordance with 4.4.3 as modified by MIL-PRF-81757/18(AS)." Delete the last sentence of subparagraph 4.5.22.c and insert "During the 60 seconds, the discharge potential shall be 7.0 ± 0.25 volts for the initial 5 seconds and 7.0 ± 0.05 volts for the remaining 55 seconds."

4.5.23 Cycling. Delete the entire text of 4.5.23 and insert the following text and table:

"4.5.23 Cycling test. Reconditioning (see 4.4.5 of MIL-PRF-81757) shall not be permitted between any of the duty cycles. This test shall not be repeated. Any noncompliance shall constitute a failure. If the battery fails to meet any one of the specified requirements of 3.10.23, as modified by MIL-PRF-81757/18(AS), at any point during the progress of the test, then terminate the test. The use of fans for additional convective cooling of the battery is permitted and encouraged. Battery temperature shall be measured at approximately the center of the battery. This test shall consist of the following steps.

"a. Preparation: Prepare the battery as follows.

"(1) Remove the battery cover.

"(2) Constant current charge the battery in accordance with 4.4.3, as modified herein.

"(3) Adjust the electrolyte levels in accordance with 4.4.4 of MIL-PRF-81757.

"(4) When performing qualification inspection, determine the battery weight.

"b. Duty cycle: Cycle the battery as follows in accordance with the parameters listed in the table below.

"(1) Discharge at 180 amperes for 4.0 minutes.

"(2) Within 4 seconds after completing step 4.5.23.b(1), constant potential charge the battery at 14.88 ± 0.1 volts for the length of time shown.

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"(3) Repeat step 4.5.23.b, continuously for the total number of cycles shown.

Category of inspection	Charge time [see step 4.5.23.b(2)]	Total # of duty cycles [see step 4.5.23.b(3)]
Qualification	116 minutes	700 duty cycles
Group C inspection	56 minutes	200 duty cycles

"c. Capacity discharge: Discharge at room temperature at 53 amperes to 9.0 volts. When performing qualification inspection, determine the battery weight.

"d. Measure the following:

"(1) The battery's current, temperature, and terminal voltage at the end of each 4-minute discharge.

"(2) The elapsed time for the battery terminal voltage to decline to 9.0 volts during the discharge of step 4.5.23.c.

"(3) The voltage of each cell at 60 minutes after the start of the discharge of step 4.5.23.d (group C inspection only).

"e. Examine the battery for the requirements of 3.10.23 as modified herein."

f. Recondition the battery in accordance with 4.4.5.

4.5.24 Qualification testing [temperature rise and float test].

4.5.24 b. Delete the text of 4.5.24.b of MIL-PRF-81757 and substitute: "Stabilize the temperature chamber that will be used for this test at 120.2 °F ± 5.4 °F (49 °C ± 3 °C)."

4.5.24.d. Delete the text of 4.5.24.d of MIL-PRF-81757 and substitute: "Immediately after the 9C-rate discharge, place the battery in the temperature chamber and charge the battery using a constant potential charge of 1.50 +0.01, -0 volts per cell for 24 ± 0.1 hours. Perform the remainder of this test with the battery in the 120.2 °F ± 5.4 °F (49 °C ± 3 °C) temperature environment."

4.5.24.e. Delete the text of 4.5.24.e of MIL-PRF-81757.

4.5.24.f. Delete the text of 4.5.24.f of MIL-PRF-81757 and substitute: "Second cycle: Discharge the battery at the 9C-rate for 3 minutes +5, -0 seconds. Continuously measure the voltage of the battery during the 9C-rate discharge. Perform step 4.5.24.1.d."

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4.5.24.g Delete the text of 4.5.24.g of MIL-PRF-81757 and substitute: "Third cycle: Repeat step 4.5.24.f."

4.5.24.h In the second sentence, delete "the C-rate" and substitute: "55 amperes".

6.13.13 Low-maintenance vented battery. Delete the first two sentences and substitute: "An aircraft battery that has a six-month maintenance interval and whose gases are vented away from the battery container (see 3.8.7 and 3.10.14.2.1 both of MIL-PRF-81757). That is, the battery will satisfactorily meet the aircraft's power and capacity requirements for not less than six-months before removal for scheduled inspection, cleaning, capacity determination, cell equalization, electrolyte adjustment, and final charge.

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TABLE III. Qualification inspection of batteries. Delete table III of MIL-PRF-81757 and insert the table shown below.

TABLE III. Qualification inspection of batteries.

Test number	Examinations and tests	Sample number					Requirement paragraph 1/	Test paragraph 1/
		1	2	3	4	5		
1	INSPECTION FOR DAMAGE	X	X	X	X		3.10.1 (G)	4.5.1 (G)
2	VISUAL AND MECHANICAL EXAM	X	X	X	X		3.10.6 (G), reqmt 4 (N) & 5 (N)	4.5.6 (G)
3	DIMENSIONS AND WEIGHT	X		X			3.10.8 (S), 3.10.9 (G), reqmt 2 (N)	4.5.8 (G)
4	CARRYING STRAP TEST			X	X		3.9 (G), 3.10.10 (G)	4.5.9 (S)
5	CELL VENT TEST	X	X	X	X		3.10.13 (G)	4.5.12 (G)
6	INITIAL CAPACITY DISCHARGE	X	X	X	X		3.10.14 (S), reqmt 3 (N)	4.5.13 (G)
7	HUMIDITY AND CHARGE RETENTION TEST			X	X		3.9 (G), 3.10.15 (G)	4.5.14 (S)
8	TEMPERATURE SHOCK TEST			X	X		3.9 (G), 3.10.17 (G)	4.5.17 (G)
9	ALTITUDE TEST			X	X		3.9 (G), 3.10.18 (G)	4.5.18 (G)
10	SALT FOG TEST	X	X				3.9 (G), 3.10.19 (G)	4.5.19 (G)
11	CONSTANT VOLTAGE DISCHARGE	X	X	X	X		3.9 (G), 3.10.22 (S)	4.5.22 (S)
12	CYCLING TEST	X	X				3.9 (G), 3.10.23 (S)	4.5.23 (S)
13	TEMPERATURE RISE AND FLOAT TEST	X	X				3.9 (G), 3.10.9 (G), 3.10.24 (G)	4.5.24 (S)
14	PHYSICAL INTEGRITY AT HIGH TEMPERATURE	X					3.9 (G), 3.10.25 (G)	4.5.25 (G)
15	INTERNAL PRESSURE TEST	X					3.9 (G), 3.10.26 (G)	4.5.26 (G)
16	CONSTANT VOLTAGE DISCHARGES		X				3.9 (G), 3.10.22 (S)	4.5.22 (S)
17	SHOCK TEST (BASIC DESIGN)			X	X		3.9 (G), 3.10.16 (G)	4.5.15 (G)
18	VIBRATION TEST			X	X		3.9 (G)	4.5.16 (G)
19	DIELECTRIC STRENGTH AND INSULATION RESISTANCE TEST	X	X	X	X		3.10.27 (G)	4.5.27 (G)
20	CHARGE STABILITY TEST (REQUIRED FOR BATTERIES WITH GAS BARRIER AND SEPARATOR SYSTEM MATERIALS NOT LISTED IN 6.10.4 (G))		X	X			3.9 (G), 3.7.11.4 (G), 3.10.24 (G), 3.10.28 (G)	4.5.28 (G)
21	STORAGE EFFECTS TEST					X	3.9 (G), 3.10.1 (G), 3.10.9 (G), 3.10.6 (G), 3.10.14 (S), 3.10.24 (G), 3.10.29 (G), reqmt 3 (N)	4.5.29 (G)

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TABLE III. Qualification inspection of batteries. (Continued)

Test number	Examinations and tests	Sample number					Requirement paragraph 1/	Test paragraph 1/
		1	2	3	4	5		
22	SHELF LIFE TEST	X			X		3.9 (G), 3.10.14 (S), 3.10.30 (G), reqmt 3 (N)	4.5.30 (G)
23	CELL BAFFLE TEST	SAMPLE CELL OF 4.2.1 (G)					3.7.11.5 (G), 3.10.12 (G)	4.5.11 (G)
24	TAMPER-RESISTANT FASTENER TEST FOR FASTENERS OTHER THAN SHEAR-HEAD SCREWS	SAMPLE CELL OF 4.2.1 (G)					3.7.9 (G)	4.5.31 (G)

1/ The notations in parentheses specify the source of the paragraph. “(G)” means the paragraph is as shown in MIL-PRF-81757. “(S)” shows that the paragraph has been modified by MIL-PRF-81757/18(AS). “(N)” indicates that a new paragraph or requirement has been added by MIL-PRF-81757/18(AS).

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TABLE IV. Qualification inspection of components and materials. Delete and substitute the following;

TABLE IV. Qualification inspection of components and materials.

Part	Examinations and tests	Requirement paragraph 1/	Test paragraph 1/
Metal parts 2/	Electrolyte resistance	3.8.1 (G), 3.8.4 (G), 3.9 (G), 3.10.2 (G)	4.5.2 (G)
Components 3/, elastomeric materials, J2 receptacle, and M81757/18-2 sensor/harness assembly	Electrolyte resistance; dielectric strength and insulation resistance	3.8.1 (G), 3.8.4 (G), 3.9 (G), 3.10.3 (S), 3.10.27 (G)	4.5.3 (S), 4.5.27 (G)
Plastic parts	Electrolyte absorption	3.8.5 (S), 3.9 (G), 3.10.4 (G)	4.5.4 (G)
Vented filler caps	Electrolyte resistance; vented filler cap test	3.9 (G), 3.10.3.1 (N), 3.10.5 (G), 3.10.6 (G)	4.5.3 (S), 4.5.5 (G), 4.5.6 (G)

1/ The notations in parentheses specify the source of the paragraph. “(G)” means the paragraph is as shown in MIL-PRF-81757. “(S)” shows that the paragraph has been modified by MIL-PRF-81757/18(AS).

2/ Including but not limited to the container, cover, and latches; external cell terminal hardware; and intercell connectors and hardware.

3/ Including but not limited to the vented filler caps, charger harnesses, battery sensor/connector assemblies, thermal switches, receptacles, heater blankets, and parts made of elastomers.

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TABLE V. Sample sizes for groups B and C testing. Delete the subgroup III column.

TABLE VII. Group B inspection. Delete table VII of MIL-PRF-81757 and substitute the table shown below.

TABLE VII. Group B inspection.

Test no.	Examinations and tests 1/	Requirement paragraph 1/	Test paragraph 1/
	Subgroup I:		
I-1	Dimensions and weight	Reqmt 2 (N), 3.10.8 (S), 3.10.9 (G)	4.5.8 (G)
I-2	Vented filler cap test (two vented filler caps per battery)	3.9, 3.10.5 (G)	4.5.5.1 (G)
	Subgroup II:		
II-1	Initial capacity discharge	Reqmt 3 (N), 3.10.14 (S)	4.5.13 (G)
	Subgroup III:		
III-1	Constant voltage discharge	3.9 (G), 3.10.22 (S)	4.5.22 (S)
III-2	Internal pressure test	3.9 (G), 3.10.26 (G)	4.5.26 (G)

1/ The notations in parentheses specify the source of the paragraph. “(G)” means the paragraph is as shown in MIL-PRF-81757. “(S)” shows that the paragraph has been modified by MIL-PRF-81757/18(AS).

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TABLE VIII. Group C inspection. Delete table VIII of MIL-PRF-81757 and insert the table given below.

TABLE VIII. Group C inspection.

Test no.	Examinations and tests	Requirement paragraph 1/	Test paragraph 1/
	Subgroup I:		
I-1	Inspection for damage	3.10.1 (G)	4.5.1 (G)
I-2	Visual and mechanical exam	Reqmt 4 (N) and 5 (N); 3.10.6 (G)	4.5.6 (G)
I-3	Dimensions and weight (weigh only after adjustment)	Reqmt 2 (N), 3.10.8 (S), 3.10.9 (G)	4.5.8 (G)
I-4	Cell vent test	3.10.13 (G)	4.5.12 (G)
I-5	Initial capacity discharge	Reqmt 3 (N), 3.10.14 (S)	4.5.13 (G)
I-6	Cycling test	3.9 (G), 3.10.23 (S)	4.5.23 (S)
I-7	Temperature rise and float test	3.9 (G), 3.10.9 (G), 3.10.24 (G)	4.5.24 (S)
	Subgroup II		
II-1	Inspection for damage	3.10.1 (G)	4.5.1 (G)
II-2	Visual and mechanical exam	Reqmt 4 (N) and 5 (N); 3.10.6 (G)	4.5.6 (G)
II-3	Dimensions and weight (weigh only after adjustment)	Reqmt 2 (N), 3.10.8 (S), 3.10.9 (G)	4.5.8 (G)
II-4	Cell vent test	3.10.13 (G)	4.5.12 (G)
II-5	Humidity and charge retention test	3.9 (G), 3.10.15 (G)	4.5.14 (S)
II-6	Constant voltage discharge	3.9 (G), 3.10.22 (S)	4.5.22 (S)

1/ The notations in parentheses specify the source of the paragraph. “(G)” means the paragraph is as shown in MIL-PRF-81757. “(S)” shows that the paragraph has been modified by MIL-PRF-81757/18(AS). “(N)” indicates that a new paragraph or requirement has been added by MIL-PRF-81757/18(AS).

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FIGURE 1. Marking for limited-repair batteries. Delete figure 1 of MIL-PRF-81757 and insert figure 4 of MIL-PRF-81757/18(AS).

FIGURE 3. Marking for battery caution and receptacle. Delete figure 3 of MIL-PRF-81757 and insert figure 5 of MIL-PRF-81757/18(AS).

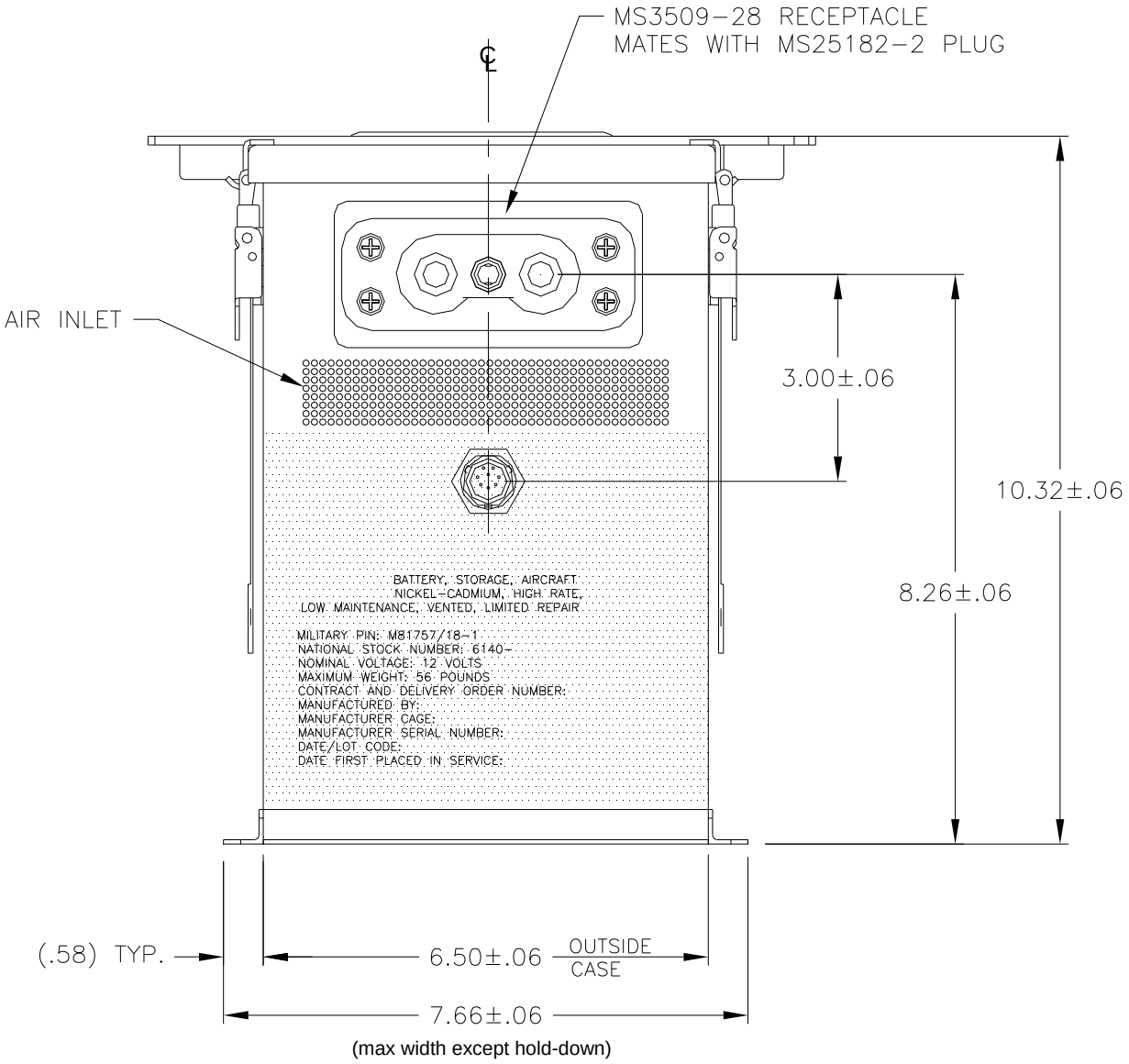


FIGURE 1. Dimensions and configuration for PIN M81757/18-1.

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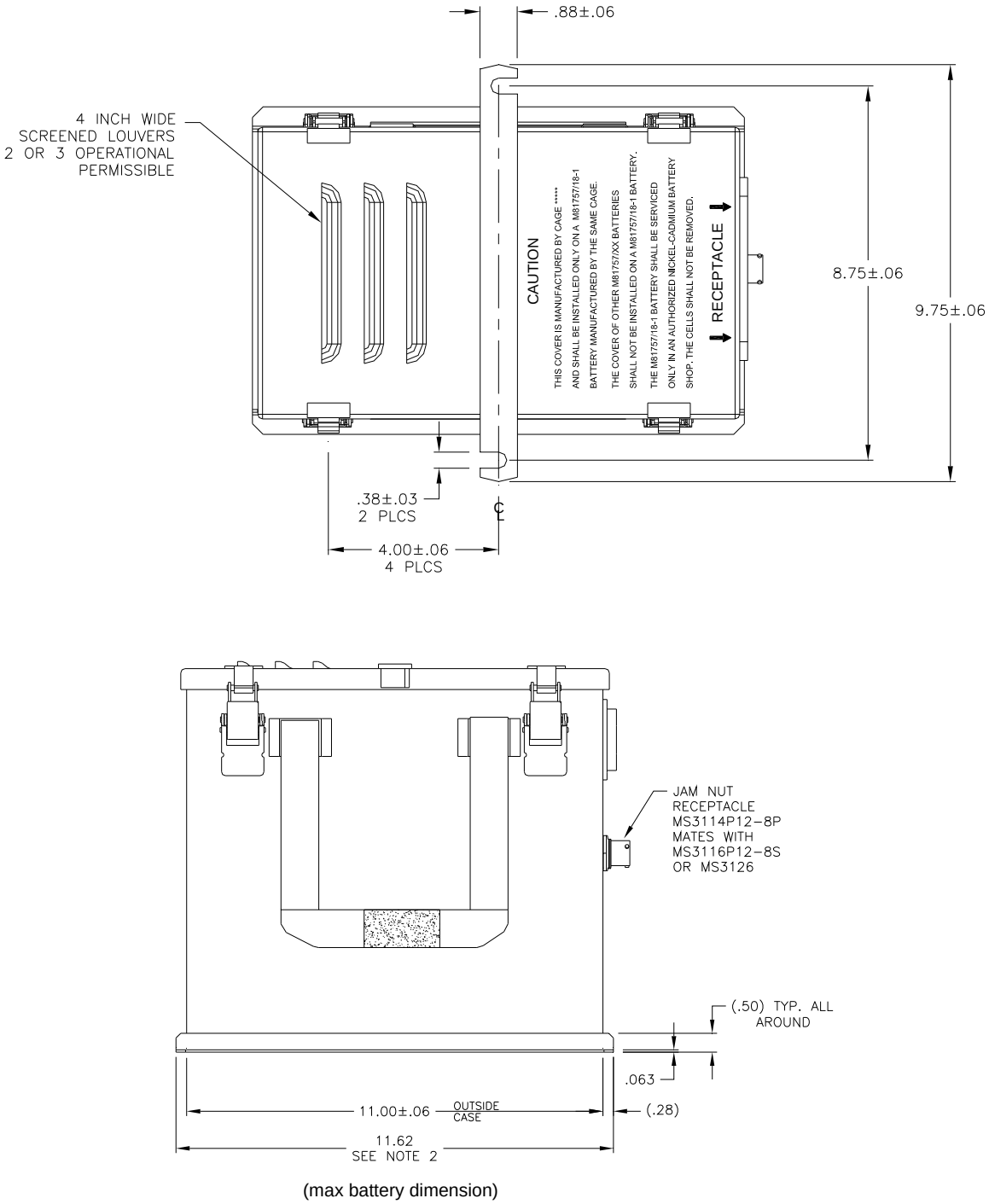


FIGURE 1. Dimensions and configuration for PIN M81757/18-1 - Continued.

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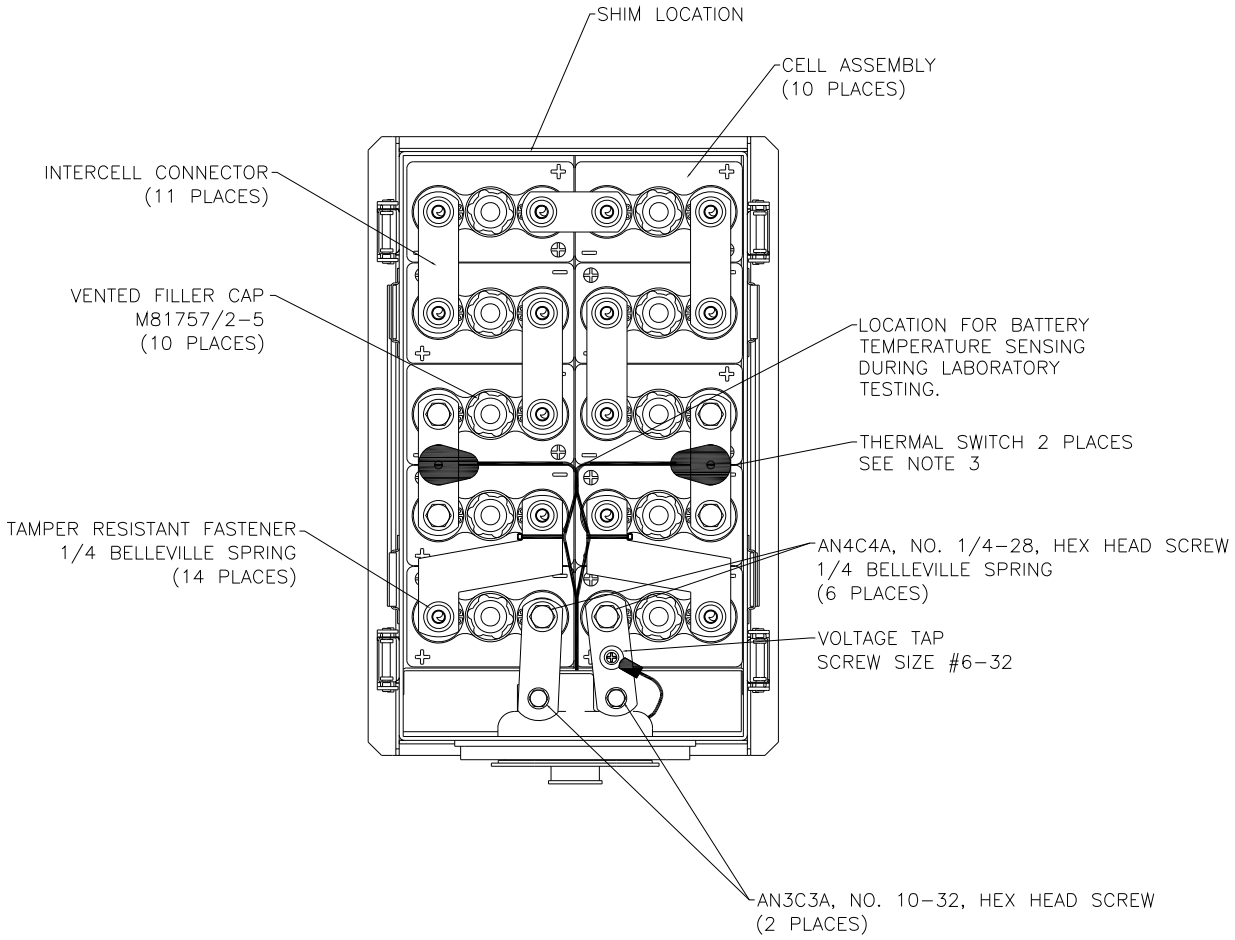
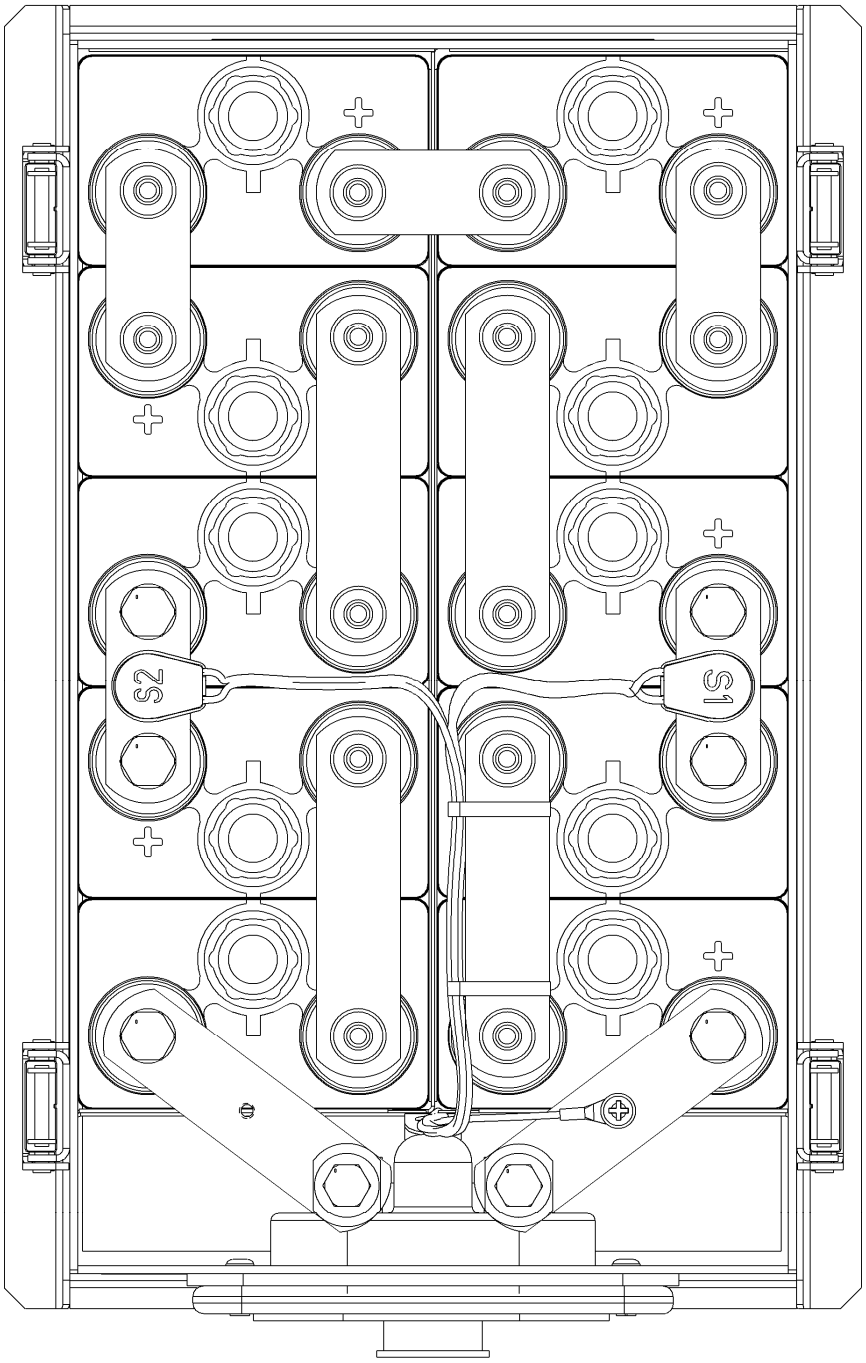


FIGURE 2. Layout (reference) of interior components for PIN M81757/18-1(design #1).

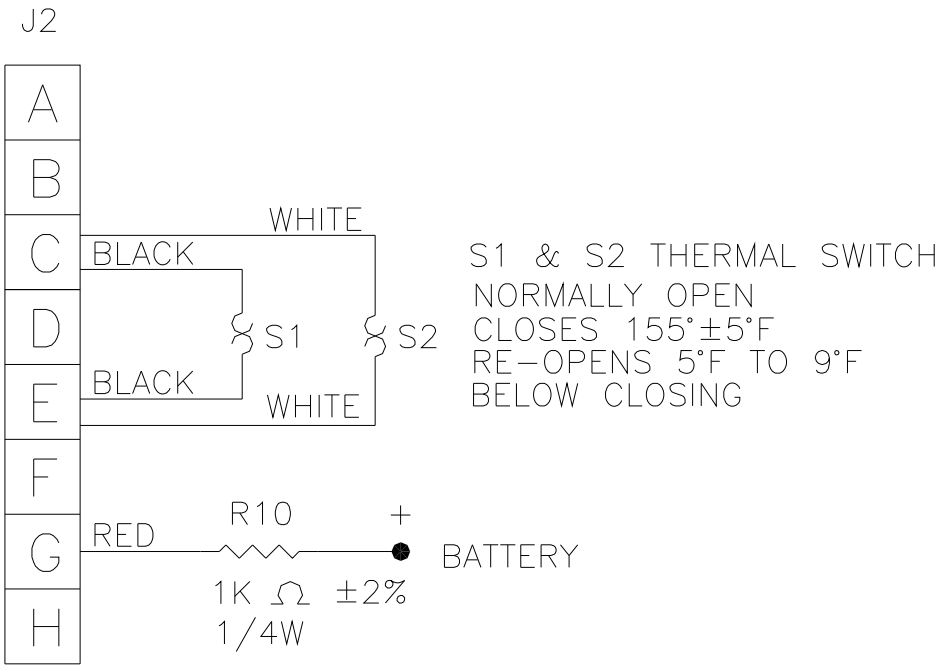
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Note #1: All notes and fastener details from design #1 apply to design #2.

FIGURE 2. Layout (reference) of interior components for PIN M81757/18-1 (design #2)-continued.

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J2 JAM NUT RECEPTACLE
(MATES WITH MS3116P128S PLUG)

SCHEMATIC

HARNESS ASSEMBLY SCHEMATIC AND PIN ASSIGNMENTS FOR M81757/18-2

Note 1: Both wires used to connect the S1 and S2 thermal switches to the J2 receptacle may be the same color (both black or both white are permitted).

FIGURE 3. M81757/18-2 sensor/harness assembly schematic.

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NOMENCLATURE:	BATTERY, STORAGE, AIRCRAFT NICKEL-CADMIUM, HIGH RATE, LOW-MAINTENANCE, VENTED, LIMITED-REPAIR
MILITARY PIN:	M81757/18-1
NATIONAL STOCK NUMBER:	6140-**-***-****
NOMINAL VOLTAGE:	12 VOLTS
MAXIMUM WEIGHT:	55 POUNDS
CONTRACT AND DELIVERY ORDER NUMBER:	
MANUFACTURED BY:	
MANUFACTURER'S CAGE:	
MANUFACTURER'S SERIAL NUMBER:	
DATE/LOT CODE:	
DATE FIRST PLACED IN SERVICE:	

FIGURE 4. Marking for M81757/18-1 batteries.

	MINIMUM LETTERING HEIGHT
CAUTION	0.500 inch minimum lettering height.
This cover manufactured by CAGE ***** and shall be installed only on an M81757/18-1 battery manufactured by same CAGE.	0.250 inch minimum lettering height.
The cover of other M81757/xx batteries shall not be installed on an M81757/18-1 battery.	0.250 inch minimum lettering height.
The M81757/18-1 battery shall be serviced only in an authorized nickel-cadmium battery shop. The cells shall not be removed.	0.250 inch minimum lettering height.
⇩ RECEPTACLE ⇩	0.500 inch minimum lettering height.

FIGURE 5. Marking for battery caution and receptacle.

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CAUTION

THIS BATTERY HAS BEEN SERVICED AS PART OF A MATCHED SET.

BOTH BATTERIES IN THE SET SHALL BE PRODUCED BY THE SAME MANUFACTURER (SEE CAGE CODE). THE BATTERIES SHALL BE PERMANENTLY PAIRED BY SERIAL NUMBER THROUGHOUT INSTALLATION, MAINTENANCE, AND CONDEMNATION.

BATTERY A: S/N _____ CAGE _____

BATTERY B: S/N _____ CAGE _____

FIGURE 6. Marking for matching battery modules.

CONCLUDING MATERIAL

Custodian:
Navy - AS

Preparing activity:
Navy - AS

Agent:
Navy - NW
(Project 6140-2007-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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